

Visual Affordance Priors for Generalizable Garment Flattening

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Abstract—Garment flattening supports downstream garment manipulation tasks such as folding and ironing. Existing value-based methods are limited by low-throughput physics, fixed-view low-fidelity rendering, and narrow asset coverage, making learned grasp values difficult to transfer to real garments with richer topology and visual variation. To address this, we develop a structural-to-visual value-learning framework spanning data generation, structural value modeling, and visual affordance distillation. First, we build an Isaac-based data pipeline that combines high-throughput physics simulation with randomized high-fidelity rendering over diverse garment assets. Second, we perform large-scale physics evaluation of dual-grasp pairs under standardized dual-arm unfolding conditions, and construct a Pair-Value Teacher to learn the structural pair-value distribution on 3D garment point clouds. Third, we project this structural value distribution onto randomized high-fidelity rendered observations and normalize it into dense image-space supervision for a directly deployable Visual Affordance Prior. Without fine-tuning on real data, our Visual Affordance Prior improves zero-shot transfer on aRTF-Clothes over ClothMate, increasing deformable/rigid Semantic Template AUC from 0.380/0.350 to 0.869/0.843. The same comparison on 22 real-robot garment assets shows higher final unfolding quality, with mean IoU increasing from 64.9 to 77.6 and coverage from 71.6 to 86.2.

I. INTRODUCTION

Before downstream garment manipulation tasks such as folding and ironing, a robot typically needs to flatten the garment [1], [2], [3]. Garment flattening often starts with dynamic unfolding, where large dual-arm motions spread out wrinkled, tangled, or highly deformed cloth, followed by quasi-static actions such as pick-and-place or local stretching to refine remaining details. We focus on the dynamic unfolding stage, where the robot must choose an effective dual-grasp pair from a single-view observation for a predefined fling primitive. This grasp-selection problem remains challenging because the garment is often severely self-occluded and geometrically complex before unfolding, while garment instances vary widely in category, silhouette, material, and texture [4].

Existing learning-based methods for this setting can be broadly grouped into two categories. One category learns policies that generate actions or trajectories directly from observations, including recent general-purpose robot policies; adapting these policies across garments, scenes, and robot embodiments often requires costly real-world adaptation, such as teleoperation demonstrations or reinforcement-learning fine-tuning [5], [6], [7]. The other category is value-based: given predefined action primitives, these methods select grasps with high predicted value instead of predicting full trajectories, thereby simplifying training and reducing

the cost of obtaining supervision from simulation [8], [1], [9], [10], [11], [12]. However, existing value-based methods are mostly developed in controlled simulation environments, such as Cloth3D assets simulated in PyFlex, where data scale, asset diversity, and visual domain randomization are limited [13], [14], [15]. Their supervision pipelines and image-space representations are also often tied to fixed viewpoints or fixed-scale assumptions. Together, these data and representation constraints limit the transfer of learned value models to real scenes. Our goal is to preserve the training advantages of value-based methods while mitigating their limitations in data generation, supervision, and visual representation for dynamic garment flattening in the real world.

To this end, we develop a structural-to-visual framework for value-based garment flattening with three components:

- 1) **Decoupled structural-to-visual data generation.** An Isaac-based pipeline decouples high-throughput physics sampling from high-fidelity rendering: the physics branch reaches about $19\times$ the rollout throughput of the original PyFlex pipeline, while the rendering branch produces observations with randomized viewpoints, lighting, materials, and scenes.
- 2) **Structural pair-value learning.** Pair-conditioned initialization and controlled directional loading measure the structural unfolding potential of ordered grasp pairs under dual-arm constraints. From sparse structural evaluations, the Pair-Value Teacher learns a pair-value distribution on normalized 3D garment point clouds, supplying structural pair-value supervision for visual transfer.
- 3) **Image-space visual affordance prior.** We convert this structural pair-value signal into normalized dense image-space targets for a Visual Affordance Prior. The resulting prior predicts a first-grasp heatmap and a conditional second-grasp heatmap, forming a deployable grasp prior for real-image inputs.

Unlike prior work that primarily evaluates value-based garment flattening in controlled benchmarks, this paper examines whether this low-cost supervision paradigm can remain effective when scaled to broader garment asset coverage, richer visual variation, and real-robot evaluation. Experiments show that the Pair-Value Teacher provides reliable structural pair-value estimates for unevaluated grasp pairs; without real-data fine-tuning, the resulting Visual Affordance Prior outperforms ClothMate on aRTF-Clothes real-image evaluation and real-robot fling experiments with 22 real garment assets.

II. RELATED WORK

A. Value-Based Cloth Unfolding

Value-based cloth unfolding methods learn action-value models that score grasp candidates from visual or geometric observations [12], [1], [9], [10], [8], [16], [11], [17]. By executing predefined open-loop primitives in simulation, they use unfolding outcomes as low-cost supervision for selecting high-value grasps, reducing cloth manipulation learning to candidate-action scoring rather than low-level control [12], [1], [8].

On the PyFlex–Cloth3D benchmark family [13], [14], [15], value-based cloth unfolding has been extensively studied across action primitives, reward design, and action-value modeling. Existing methods cover static pick-and-place, pick-and-stretch, dynamic fling, and their combinations [12], [9], [8]. Reward definitions have progressed from coverage rewards to canonicalized-alignment objectives and orientation rewards, extending evaluation from flattening alone to alignment and orientation [12], [1], [17]. In terms of model design, spatial action maps have been extended with Transformer visual encoders, 3D geometry, or keypoint priors for stronger candidate scoring [12], [1], [11], [10]. ClothMate further exploits Grasp-Fling Consistency to decouple structural learning from visual policy learning, achieving strong benchmark performance with only about 15% of the original training data and reaching 91.5% coverage in 4.7 steps on average [8].

These advances demonstrate the effectiveness of value-based supervision for learning cloth grasp quality. However, these methods were primarily developed and evaluated within controlled PyFlex–Cloth3D benchmarks, whose assumptions do not fully match the requirements of real-world deployment. The benchmark environments provide limited simulation throughput, visual realism, and asset coverage, and many methods built on them design supervision and representations around fixed views and scale-dependent objectives [13], [14], [15]. Our work keeps the low-cost supervision advantage of value-based learning, but rebuilds the data-generation pipeline in Isaac and adapts the supervision construction and visual representation so that value-based supervision can transfer to image-space affordance prediction under broader topology and visual variation.

B. Synthetic Data and Cloth Simulation

Simulation-generated data has proved effective for real-world transfer in visual and robotic learning when synthetic assets, photorealistic rendering, domain randomization, and simulated interaction cover sufficient variation in assets, textures, lighting, viewpoints, and scenes [18], [19], [20], [21]. For value-based cloth learning, this motivates rebuilding the data-generation pipeline to support high-fidelity rendering and richer randomization, rather than only action-value evaluation in controlled benchmarks. Recent sim-to-real grasping systems further show that teachers learned from low-cost simulation or privileged structural information can be distilled into visual models, reducing the burden

of learning task-relevant supervision from expensive visual observations [22], [23]. We follow this structural-to-visual separation within ClothMate’s two-stage value-learning formulation [8]: high-throughput physics-only simulation learns structural pair value in a normalized domain, while a smaller but visually richer collection of rendered observations distills this knowledge into image space.

Cloth simulation has advanced in efficiency, physical fidelity, asset scale, and task coverage, from PBD and FEM-based methods to environments such as GarmentLab and DexGarmentLab [24], [25], [26], [27]. Yet real garments remain difficult to calibrate at scale in material properties, friction, bending, stretching, and contact dynamics; even RGBench, with system identification and a high-fidelity scalable simulator, reports residual dynamic simulation error [27]. We therefore avoid defining grasp-pair value by full dynamic fling outcomes from random initial states. Instead, we reformulate value measurement as structural unfolding-potential evaluation under pair-conditioned initialization and controlled directional loading, producing supervision that better reflects pair-specific unfolding potential while reducing incidental effects from random wrinkles, tangling, or specific fling parameters.

III. METHOD

A. Method Overview

Given a single-view RGB observation from an arbitrary viewpoint, our goal is to select an ordered grasp pair on the visible garment region for dual-arm dynamic unfolding. To combine scalable structural supervision with deployable image-space prediction, we structure the learning pipeline around structural-to-visual pair-value transfer: a structural pair-value field is first learned on canonicalized 3D garment point clouds and then projected into image-space affordance supervision for RGB-based deployment.

As shown in Fig. 1, the pipeline consists of three connected components:

- 1) **Dual-branch data generation (Sec. III-B)** decouples high-throughput physical sampling from randomized high-fidelity rendering over a shared garment asset pool, producing structural pair-value labels and image observations with mesh correspondences.
- 2) **Structural pair-value label generation and supervision (Secs. III-C–III-D)** generates sparse labels through pair-conditioned initialization, controlled directional loading, and an adapted canonicalized-alignment reward, and then uses them to provide structural supervision over unevaluated grasp pairs.
- 3) **Image-space affordance supervision (Sec. III-E)** projects the learned pair-value field into the visible image region through mesh correspondence and barycentric interpolation, producing heatmap supervision for a Visual Affordance Prior that operates on RGB images.

B. Dual-Branch Data Generation

The Pair-Value Teacher and Visual Affordance Prior require different forms of data: large-scale pair-level physics

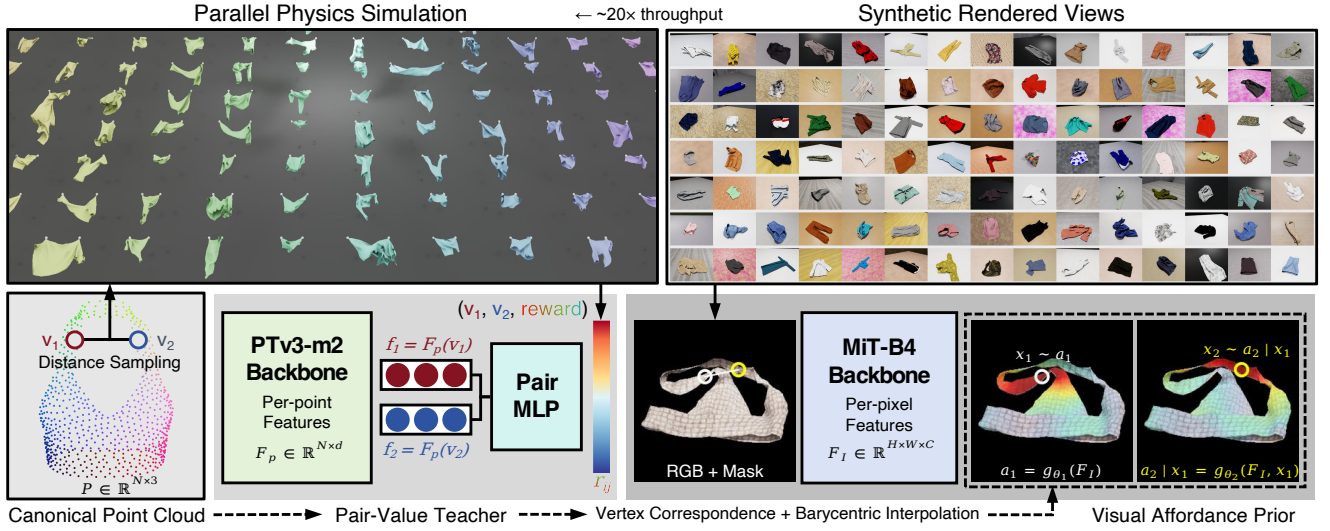


Fig. 1. **Overview of the structural-to-visual affordance learning pipeline.** Stage I collects large-scale (asset, pair, reward) labels with Parallel Physics Simulation and trains a Pair-Value Teacher on canonical point clouds. Stage II projects Pair-Value Teacher predictions into rendered observations through vertex correspondence and barycentric interpolation to supervise the Visual Affordance Prior.

evaluations for structural value learning, and randomized high-fidelity image observations for real-image transfer. Because physics-only sampling is much cheaper than high-fidelity rendering, we decouple structural supervision collection from visual observation generation into two branches over a shared garment asset pool: Parallel Physics Simulation and Synthetic Rendered Views. The two branches preserve mesh correspondences, allowing structural pair values to be projected into image-space supervision. Starting from the ClothesNet assets filtered by DexGarmentLab [26], we further remove garments that are unsuitable for flattening and retain 1,646 garments.

Parallel Physics Simulation uses PhysX XPBD to run parallel cloth simulations for high-throughput evaluation of ordered grasp pairs. This branch samples ordered grasp pairs and evaluates them with pair-conditioned initialization and controlled directional loading (Sec. III-C), producing structural pair-value labels. The Isaac-based rollout pipeline achieves about 15.5k action samples per hour, compared with about 800 samples per hour in the original PyFlex pipeline. This higher throughput allows structural supervision to cover a larger and more diverse asset pool.

Synthetic Rendered Views generates randomized high-fidelity visual observations from the same asset pool for subsequent visual distillation. Each garment is first initialized above the workspace in a random pose and allowed to settle under gravity, yielding diverse folded and wrinkled configurations. We then randomize camera viewpoints and intrinsics, lighting, backgrounds, ground materials, and garment appearances, using both native ClothesNet textures and MatSynth PBR fabric materials [28] to increase variation in viewpoint, geometry, and appearance. The rendered data retain image-to-mesh correspondences, which later enable structural pair values to be projected into image space.

C. Structural Pair-Value Label Generation

To generate structural supervision for the Pair-Value Teacher, we assign each ordered grasp pair a structural pair value through three steps. Pair-conditioned initialization constructs a pair-specific initial state, controlled directional loading unfolds the garment under dual-grasp constraints to yield a final cloth state, and reward formulation converts this state into the pair’s structural pair value.

Pair-conditioned initialization. Given an ordered grasp pair, the system starts from a canonical garment state and applies a rigid transformation that places the selected pair at the dual-arm grasp poses while aligning the principal garment direction with the prescribed unfolding direction. This pair-centered setup makes the reward reflect the pair’s structural effect rather than incidental wrinkles or self-entanglement from a random initial state.

Controlled directional loading. As illustrated by the Cond-D sequence in Fig. 2, after pair-conditioned initialization, the two grasp points are pulled apart to tension the cloth between the grippers. Gravity is then disabled and a uniform external force is applied along the prescribed loading direction; after a short settling period, gravity is restored and the garment is released. This controlled directional-loading procedure reduces dependence on dynamic parameters such as fling speed and amplitude while producing the final state for reward formulation.

Reward formulation. We map the final cloth state to a scalar structural pair value using an adapted canonicalized-alignment reward from Cloth-Funnels [1]. Following Cloth-Funnels, the reward factorizes vertex-wise distance to a canonical flat reference into deformable and rigid distances through ICP-style alignment. We allow the rigid alignment to translate freely along the loading direction, preventing action-induced displacement from biasing the grasp-pair value. The adapted reward is used as the structural pair value.

for all subsequent image-space supervision.

D. Structural Pair-Value Supervision

Exhaustive simulation over all candidate grasp pairs is impractical because the pair space grows quadratically with the number of sampled surface points. We therefore construct a Pair-Value Teacher to estimate structural pair values for unevaluated candidate pairs from sparse structural pair-value labels. Given a normalized garment point cloud $\mathcal{P}$ and its point normals $\mathcal{N}$, the model predicts the value of an ordered candidate vertex pair as

$$\mathbf{F} = \Phi_\theta(\mathcal{P}, \mathcal{N}), \quad T_\theta(v_a, v_b) = g_\theta(\mathbf{F}_{v_a}, \mathbf{F}_{v_b}), \quad (1)$$

where T_θ denotes the prediction function of the Pair-Value Teacher, Φ_θ is the PTv3-m2 backbone [29], g_θ is the pair MLP, and $\mathbf{F}_{v_a}$ and $\mathbf{F}_{v_b}$ are the point features at the ordered candidate garment vertices v_a and v_b . The Pair-Value Teacher is trained with a Huber loss against the structural pair-value labels generated in Sec. III-C.

Unlike Spatial Action Maps, which encode grasp-pair values after projecting the garment surface to a 2D domain, the Pair-Value Teacher operates directly on the 3D point cloud. This representation preserves surface adjacency, avoids projection discontinuities near seams, and reduces detail loss from garment-scale variation and finite 2D grid resolution.

E. Image-Space Affordance Supervision

Given the structural pair values predicted by the Pair-Value Teacher, we project them into image space through **Vertex Correspondence** and **Barycentric Interpolation**. For a foreground pixel x , rasterization returns the three garment mesh vertices $\{v_m(x)\}_{m=1}^3$ of the visible triangle covering x and their barycentric coordinates $\lambda(x) = \{\lambda_m(x)\}_{m=1}^3$. For a pixel pair (x_1, x_2) , we evaluate the 3×3 vertex-pair values between the two visible triangles and interpolate them into a pixel-pair supervision signal:

$$S(x_1, x_2) = \sum_{m,n=1}^3 \lambda_m(x_1) \lambda_n(x_2) T_\theta(v_m(x_1), v_n(x_2)). \quad (2)$$

From the projected pair value $S(x_1, x_2)$, we construct **Normalized Conditional Heatmaps** as supervision for the Visual Affordance Prior. Let Ω denote the visible garment region, $\mu_k(\cdot)$ denote the top- k mean operator, and $\mathcal{E}_{x \in \Omega}(\cdot)$ denote masked exponential normalization over the variable x within Ω . The first-grasp and conditional second-grasp heatmap targets are defined as

$$\begin{aligned} A_1^*(x_1) &= \mathcal{E}_{x_1 \in \Omega}(\mu_k\{S(x_1, x_2)\}_{x_2 \in \Omega}), \\ A_2^*(x_2 | x_1) &= \mathcal{E}_{x_2 \in \Omega}(S(x_1, x_2)). \end{aligned} \quad (3)$$

Here, $A_1^*(x_1)$ measures the potential of x_1 as a first grasp that can form high-value pairs, while $A_2^*(x_2 | x_1)$ gives the relative affordance of x_2 as the second grasp conditioned on x_1 . The Visual Affordance Prior uses a SegFormer-MiT-B4 backbone [30] to extract visual features from a masked RGB image. It first predicts $\hat{A}_1(x)$ and then, given a sampled

first-grasp query x_1 , predicts the query-conditioned heatmap $\hat{A}_2(x_2 | x_1)$. We train the model with a masked KL loss against the corresponding targets A_1^* and A_2^* .

This normalized heatmap formulation differs from Spatial Action Maps, which directly regress canonicalized-alignment rewards over pixel-action grids with discrete rotations, scales, and positional encodings [31], [1], [8]. Under randomized viewpoints, the relation between image-space action scale and physical grasp scale becomes unstable. Normalizing pair values within each observation instead encourages the visual model to learn relative grasp affordance in the current view, rather than raw rewards tied to physical scale.

IV. EXPERIMENTS

A. Experimental Setup

Data generation and splits. Experiments are built on a shared pool of 1,646 garment assets. Structural supervision is generated with Parallel Physics Simulation: for each garment, 128 anchor points are selected from the template point cloud using FPS, and 512 grasp-point pairs are sampled from distance-stratified anchor pairs for simulation-based evaluation. The resulting pairwise structural labels are split at the pair level within the same asset pool, yielding 757,160 training labels and 85,592 validation labels. Visual supervision is generated from Synthetic Rendered Views and split at the asset level, producing 25,920 training images and 3,708 validation images. Rendering randomizes camera pose, lighting, background, and material appearance, using both native ClothesNet textures and 385 MatSynth PBR fabric materials.

Model training and real-world deployment. The Pair-Value Teacher is trained with AdamW for 100 epochs using batch size 32, learning rate 10^{-3} , and weight decay 0.05. For the Visual Affordance Prior, input images are resized to 320×256 and normalized with ImageNet statistics. The target A_1^* is computed as the mean value of the top-8 candidate second grasp points in $S(x_1, \cdot)$, and each image samples 4 first-grasp queries to supervise the corresponding A_2^* . Both heatmap targets use masked exponential normalization with temperature $\tau = 0.1$ and are optimized with equally weighted masked KL losses. The visual model is trained with AdamW for up to 100 epochs using batch size 16, learning rate 10^{-4} , and weight decay 10^{-4} . Garment masks are obtained directly from rendering for synthetic images, and from SAM3 with text prompts for real images and robot experiments.

B. Structural Supervision Protocol

This experiment evaluates the effect of pair-conditioned initialization and controlled directional loading on structural pair-value label generation. We combine random or pair-conditioned initialization with either fixed-parameter fling or controlled directional loading, yielding four offline supervision protocols: Rand-F, Rand-D, Cond-F, and Cond-D. We report the mean coverage with respect to a canonical flat reference and the standard deviation of coverage across repeated trials, which measure unfolding performance

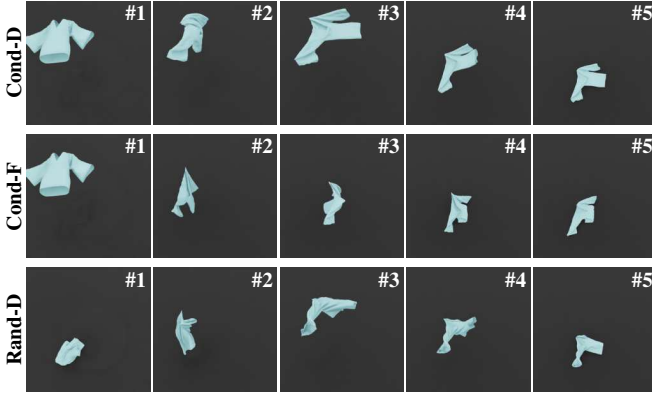


Fig. 2. **Representative offline-supervision sequences.** Columns #1–#5 show the evaluation process of the same grasp pair under Cond-D, Cond-F, and Rand-D. Cond-D produces a more complete unfolding sequence, while Rand-D can suffer from tangling caused by the random initial state and Cond-F can leave the lower garment folded when the fixed-parameter fling has insufficient speed or amplitude.

and label stability, respectively. The experiment covers 100 garments, with 64 ordered grasp-point pairs sampled per garment. Each protocol is repeated 8 times per pair, resulting in 51,200 simulated trials in total.

TABLE I
COMPARISON OF OFFLINE SUPERVISION PROTOCOLS.

Protocol	Cov. Mean (%) $\uparrow$				Cov. Std (%) $\downarrow$
	Low	Mid	High	All	All
Rand-F	73.6	77.6	76.2	75.8	8.4
Rand-D	74.6	82.4	84.6	80.5	8.4
Cond-F	75.1	77.9	76.7	76.5	3.4
Cond-D	78.5	86.4	89.3	84.7	2.6

Tab. I reports quantitative results for the four protocols across garments of different mesh complexity, where Low, Mid, and High are defined by the number of mesh vertices. Pair-conditioned initialization leads to more stable repeated evaluations: under fixed-parameter fling, the coverage standard deviation decreases from 8.4 for Rand-F to 3.4 for Cond-F; under controlled directional loading, it decreases from 8.4 for Rand-D to 2.6 for Cond-D. Controlled directional loading also achieves higher mean coverage, improving from 75.8 to 80.5 under random initialization and from 76.5 to 84.7 under pair-conditioned initialization. This advantage is more pronounced for High-complexity garments. Fig. 2 illustrates two representative failure modes: Rand-D is susceptible to random initial entanglement, leading to insufficient unfolding, whereas Cond-F may leave residual folds in the lower part of the garment when the velocity or displacement amplitude of the fixed-parameter fling is too small.

C. Pair-Value Teacher Generalization

This experiment evaluates whether the Pair-Value Teacher can provide reliable structural pair-value estimates for un-evaluated grasp pairs. For each garment, the observed pool

contains 512 grasp pairs evaluated in simulation. We construct a disjoint unseen-pair pool and score its candidates with the Pair-Value Teacher. We then execute 64 unseen pairs per garment in simulation: 16 high-scoring candidates are used to test whether selecting from the unseen pool can improve over the best evaluated pair in the observed pool, while another 48 candidates are stratified by grasp-point distance and predicted score to evaluate prediction error and ranking consistency. The experiment covers 100 garments and executes 6,400 unseen pairs in total.

TABLE II
UNSEEN-PAIR RETRIEVAL RESULTS OF THE PAIR-VALUE TEACHER.

Metric	Value
Observed-pool best reward $\uparrow$	-0.0734
Retrieved unseen best reward $\uparrow$	-0.0549
Improved assets $\uparrow$	85/100
Reward MAE $\downarrow$	0.0183
Spearman’s ρ $\uparrow$	0.965

Tab. II reports the execution results. The high-scoring unseen candidates outperform the observed-pool best on 85/100 garments. Averaged across garments, the observed-pool best reward is -0.0734 , whereas the best retrieved unseen reward is -0.0549 . For the stratified unseen candidates, the Pair-Value Teacher achieves a reward MAE of 0.0183 against executed rewards, and the mean per-garment Spearman’s ρ reaches 0.965, indicating strong rank consistency between predicted scores and executed rewards. These results suggest that the Pair-Value Teacher provides accurate structural pair-value estimates for unseen grasp pairs and guides search toward better grasp-pair candidates.

D. Visual Affordance Transfer

We evaluate zero-shot visual affordance transfer on real images from aRTF-Clothes, using the publicly released ClothMate checkpoint as the baseline. Since aRTF provides semantic keypoints rather than physical execution outcomes, we measure whether a predicted grasp pair matches a valid semantic support pattern. For an image n of category c_n , let $\mathcal{K}_{c_n}$ be its annotated semantic keypoints and $\mathcal{T}_{c_n}$ be a category-specific set of valid keypoint-pair templates. Given a predicted grasp pair $(p_{n,1}, p_{n,2})$, let $\hat{k}_{n,i}$ denote the nearest semantic keypoint to $p_{n,i}$ and $z_n = \mathbb{I}[(\hat{k}_{n,1}, \hat{k}_{n,2}) \in \mathcal{T}_{c_n}]$. We define

$$d_n = \sqrt{\frac{1}{2} \sum_{i=1}^2 \left(\frac{\min_{k \in \mathcal{K}_{c_n}} \|p_{n,i} - k\|_2}{s_{n,\text{box}}} \right)^2}. \quad (4)$$

$$\text{STAUC} = \int_0^1 \frac{1}{N} \sum_{n=1}^N z_n \mathbb{I}[d_n \leq \tau] d\tau. \quad (5)$$

where $s_{n,\text{box}}$ is the instance bounding-box scale. Deformable and rigid grasps follow the same definition but use different template sets: deformable templates include semantically valid local or global opening supports under garment shape variation, while rigid templates retain stricter global support

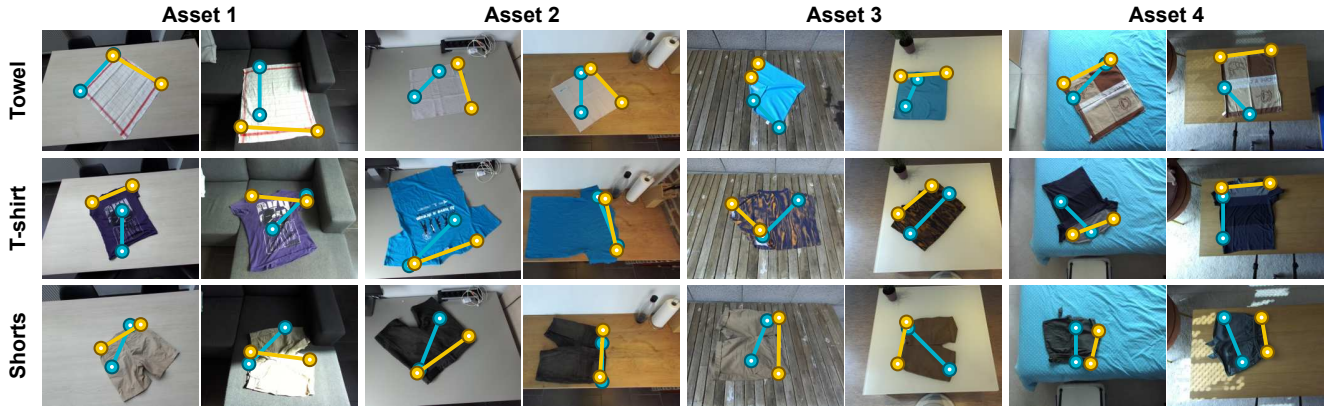


Fig. 3. **Qualitative comparison on aRTF-Clothes across scene and asset variations.** Rows correspond to towel, T-shirt, and shorts categories, and the four two-column groups show Assets 1–4 under two observations each. Each panel overlays grasp pairs predicted by ClothMate (—) and the Visual Affordance Prior (—) on the same real image.

TABLE III
SEMANTIC TEMPLATE AUC ON ARTF-CLOTHES.

Category	Deformable $\uparrow$		Rigid $\uparrow$	
	ClothMate	Ours	ClothMate	Ours
Towel	0.677	0.855	0.677	0.855
T-shirt	0.155	0.909	0.144	0.889
Shorts	0.226	0.811	0.089	0.720
Overall	0.380	0.869	0.350	0.843

poses. As shown in Tab. III, over the full dataset, our deformable/rigid scores are 0.869/0.843, compared with 0.380/0.350 for ClothMate.

Fig. 3 compares qualitative predictions across paired observations of the same garment assets. Across changes in lighting and viewpoint, our predictions usually remain on similar, category-relevant support structures, such as shoulder-side grasps for T-shirts, waistband-side grasps for shorts, and same-side corner grasps for towels. In contrast, ClothMate shows larger shifts across observations and is sometimes attracted to texture patterns or visually salient but structurally implausible regions. The trend is not absolute: for shorts, our prediction can occasionally move toward a same-side waistband–leg pair, and under uneven illumination, such as Shorts/Asset-1/Sample-2, it can drift from the desired support.

E. Real-Robot Fling Evaluation

We further evaluate the effect of the Visual Affordance Prior on real-world unfolding performance, using ClothMate as the baseline. As shown in Fig. 4, the setup consists of two Diana-7 arms, Pika grippers, and a front-mounted D435i RGB-D camera. Both methods use the same execution protocol. After filtering out unreachable candidates, the robot moves downward under impedance control to grasp the selected pair, uses visual feedback during Lift and Stretch to check table clearance and cloth tautness, and then performs the dynamic fling.

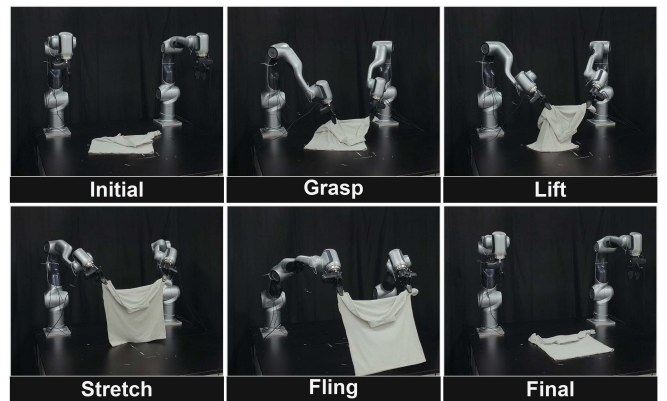


Fig. 4. **Real-robot execution protocol.** Keyframes show the sequence from the initial configuration through grasping, lifting, stretching, and fling execution to the final state.

Fig. 5 shows the 22 real garment assets used in the evaluation. Each asset is evaluated with five trials and a budget of three fling actions per trial. We record only the final unfolded state of each trial, and compute IoU and coverage (Cov.) with respect to the standard flattened state of the corresponding garment. Tab. IV reports the final-step results for each asset, with each value shown as the mean and the half-width of the bootstrap 95% confidence interval. Overall, our method improves the asset-level average from 64.9 to 77.6 in IoU and from 71.6 to 86.2 in coverage.

Fig. 6 shows representative real-robot trials. Qualitatively, our method typically produces better unfolding outcomes on garments with more consistent texture and relatively stable topology, such as pants and upper-body garments. For long skirts, however, inside-out flipping can lead to failure. Irregular textures can also affect the prediction, where distinctive stripe boundaries may be mistaken as meaningful grasp locations.

V. CONCLUSION

Contributions. We presented a pair-conditioned learning framework for garment flattening that turns dual-grasp super-

TABLE IV
REAL-ROBOT FLING METRICS ON 22 GARMENT ASSETS.

Metric	Method	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11
IoU	ClothMate	71.7±11.6	76.5±5.4	65.7±9.4	72.4±7.0	73.2±9.6	69.0±12.8	49.9±5.5	71.1±14.4	67.3±7.7	74.8±7.3	56.0±10.7
	Ours	84.0±4.9	85.7±4.2	83.3±5.9	88.9±0.9	78.7±7.8	86.4±1.8	75.8±7.4	82.3±14.3	83.0±5.0	88.7±2.4	76.9±2.9
Cov.	ClothMate	75.6±11.2	81.0±5.8	70.6±9.1	74.6±7.8	77.9±10.6	74.4±15.4	55.3±8.0	73.4±15.1	69.3±9.2	76.7±6.9	60.2±11.3
	Ours	89.3±5.1	90.4±4.9	87.7±5.4	92.7±1.7	91.2±10.4	95.5±3.0	82.8±8.6	83.6±14.1	87.6±6.9	90.6±3.1	89.5±2.7
Metric	Method	#12	#13	#14	#15	#16	#17	#18	#19	#20	#21	#22
IoU	ClothMate	67.0±6.8	56.0±19.5	61.5±5.5	62.0±7.7	56.8±22.1	66.9±15.5	63.5±5.9	55.3±3.8	54.6±7.2	71.2±3.2	65.9±7.4
	Ours	70.0±5.4	61.3±8.3	68.7±12.3	51.7±6.7	79.7±10.7	88.3±5.2	88.7±2.5	58.2±11.8	80.2±2.5	79.8±5.3	67.2±13.0
Cov.	ClothMate	71.2±6.3	62.2±23.3	93.2±4.1	74.6±7.9	69.2±25.2	72.3±14.4	69.6±7.0	61.8±4.6	63.9±9.7	76.0±4.1	71.2±7.3
	Ours	75.3±4.6	74.7±14.9	85.1±4.6	68.9±9.0	93.3±6.3	96.4±3.5	94.7±1.4	67.2±10.1	91.7±2.1	88.4±4.5	78.6±14.1



Fig. 5. Real garment assets used in the real-robot fling evaluation. Asset numbers correspond to the columns in Tab. IV.

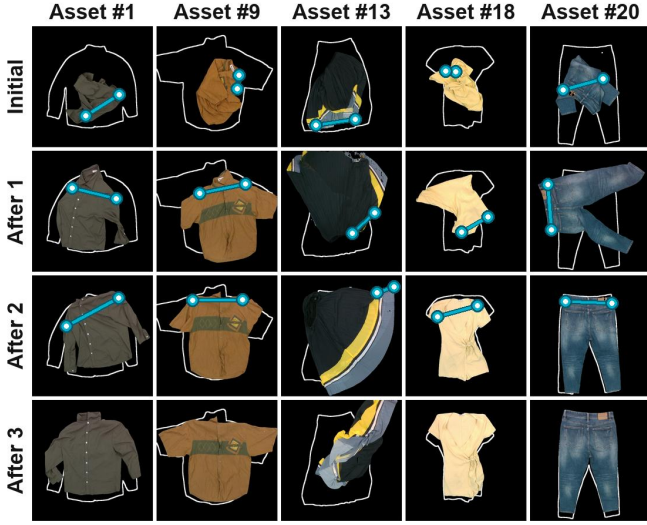


Fig. 6. **Representative real-robot qualitative results.** Each column shows one garment asset from the initial observation through three fling actions. The white contour indicates the flattened target silhouette, and cyan markers denote the selected grasp pairs.

vision into pair-quality estimation and distills the resulting structural knowledge into an image-space visual affordance prior. Experiments show that the proposed protocol yields the most stable supervision, that the teacher retrieves better unseen grasp pairs, and that the distilled visual prior transfers zero-shot to real garment images with semantically meaningful behavior. Real-robot fling evaluation further indicates that the learned prior improves final unfolding quality over the ClothMate baseline under a fixed flattened-target metric.

Limitations. The deterministic heatmap prior can still fa-

vor dominant support modes, and despite texture and lighting randomization, high-contrast patterns or uneven illumination may pull predictions toward spurious visual cues. On the robot, the embodiment-agnostic prior is paired with an open-loop fling primitive, so reachability filtering, grasp failures, and primitive-parameter mismatch remain limiting factors; future work should integrate robot-aware multimodal grasp selection with closed-loop execution.

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